DIODE CHARACTERISTIC CURVE

SEMICONDUCTOR DIODE

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Diode Approximations

Ideal, Second, and Third Approximation

Shockley's Equation

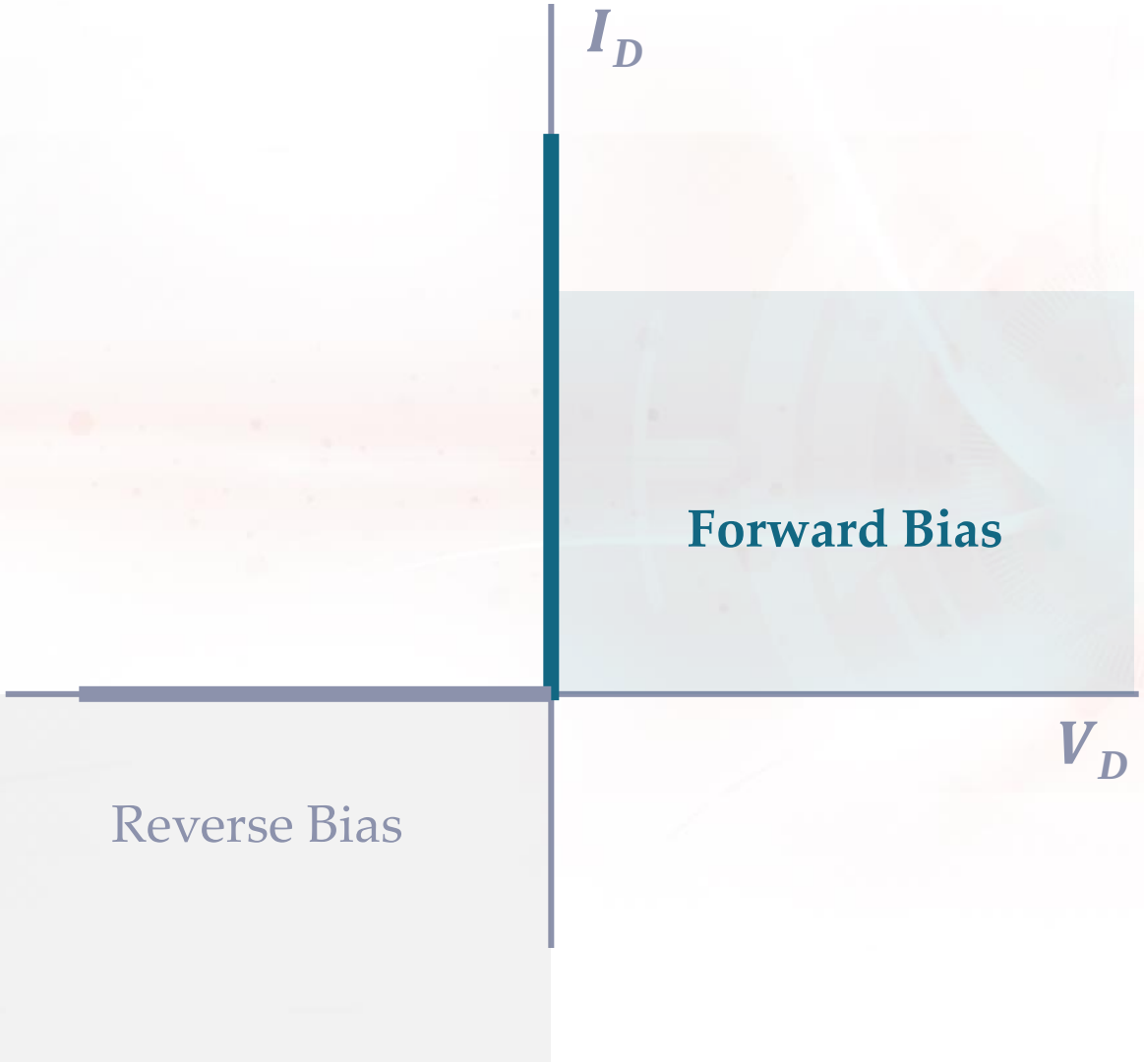
Load Line Analysis

Reading Data Sheet

DIODE APPROXIMATIONS

IDEAL DIODE

Characteristic Curve



Acts like a **switch**:

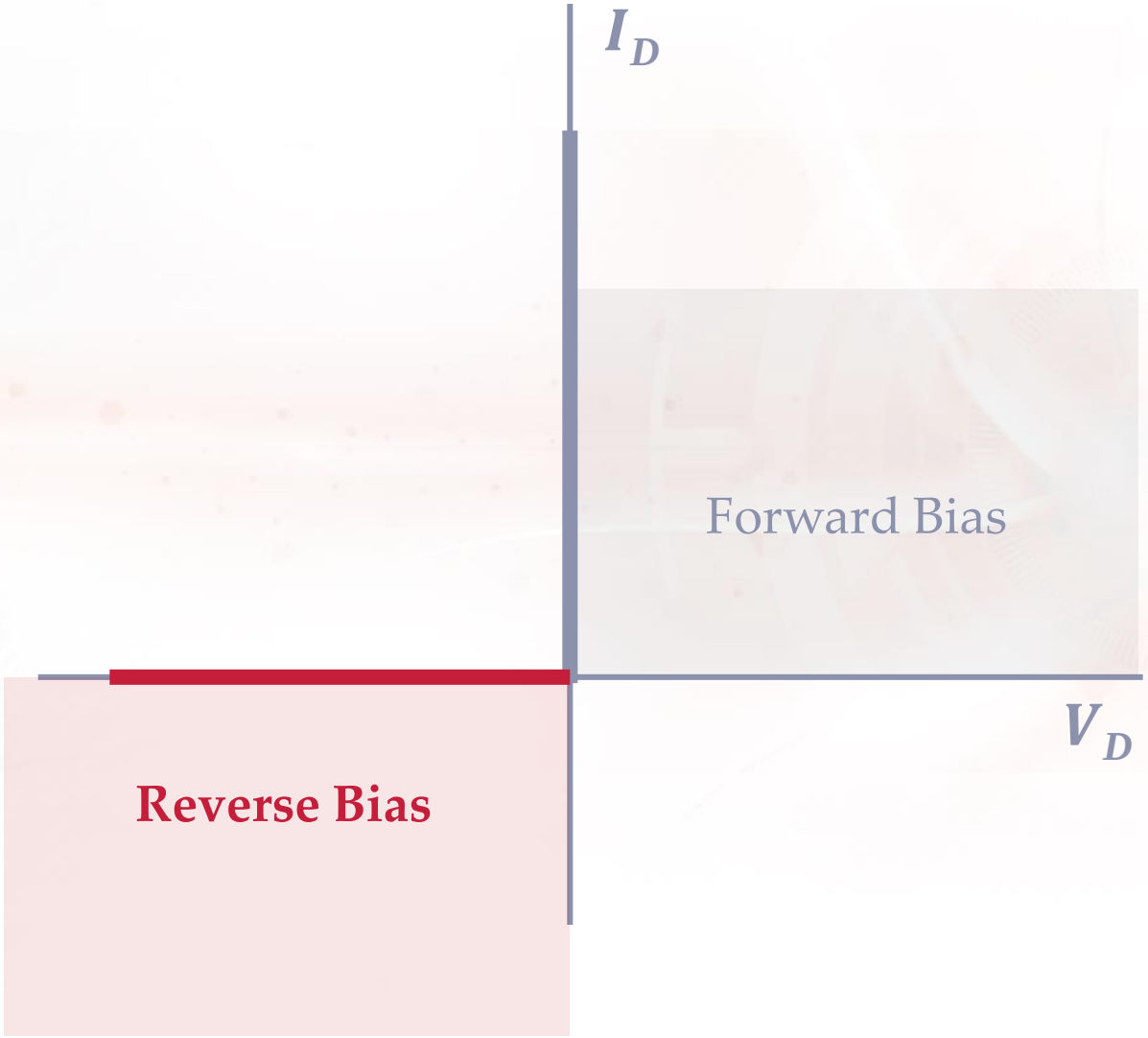
Zero (0) resistance when **forward** bias

Circuit Equivalent

Forward Bias

IDEAL DIODE

Characteristic Curve



Acts like a **switch**:

Infinite (∞)resistance when **reverse** bias

Circuit Equivalent

Reverse Bias

Use the **ideal** diode to calculate the load **voltage** and load **current** of the given network.

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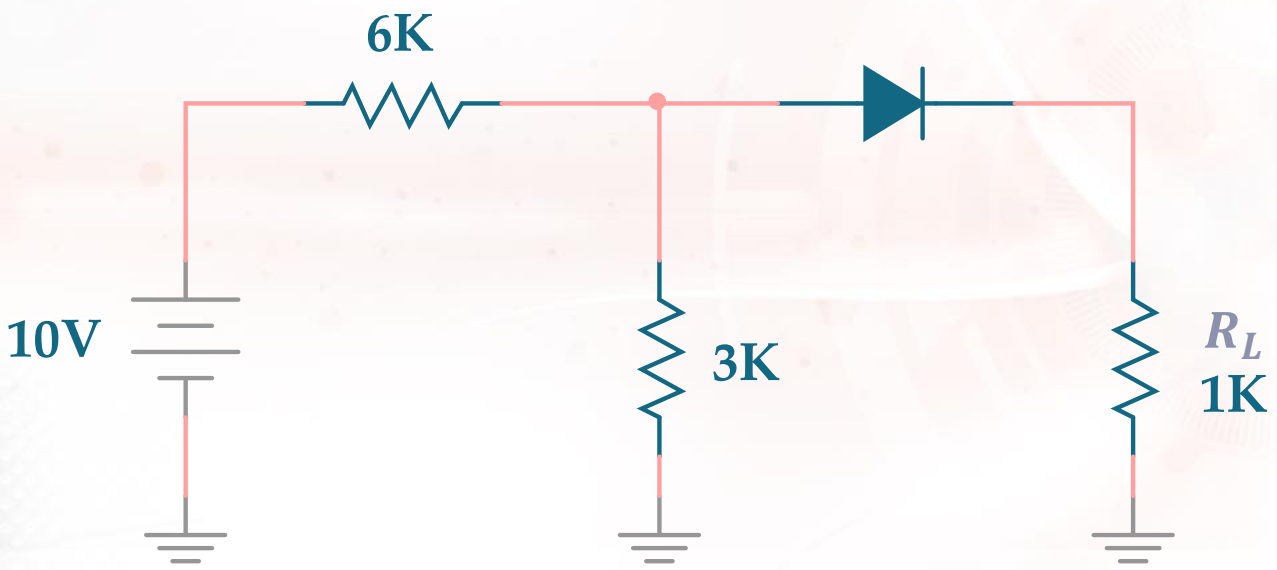
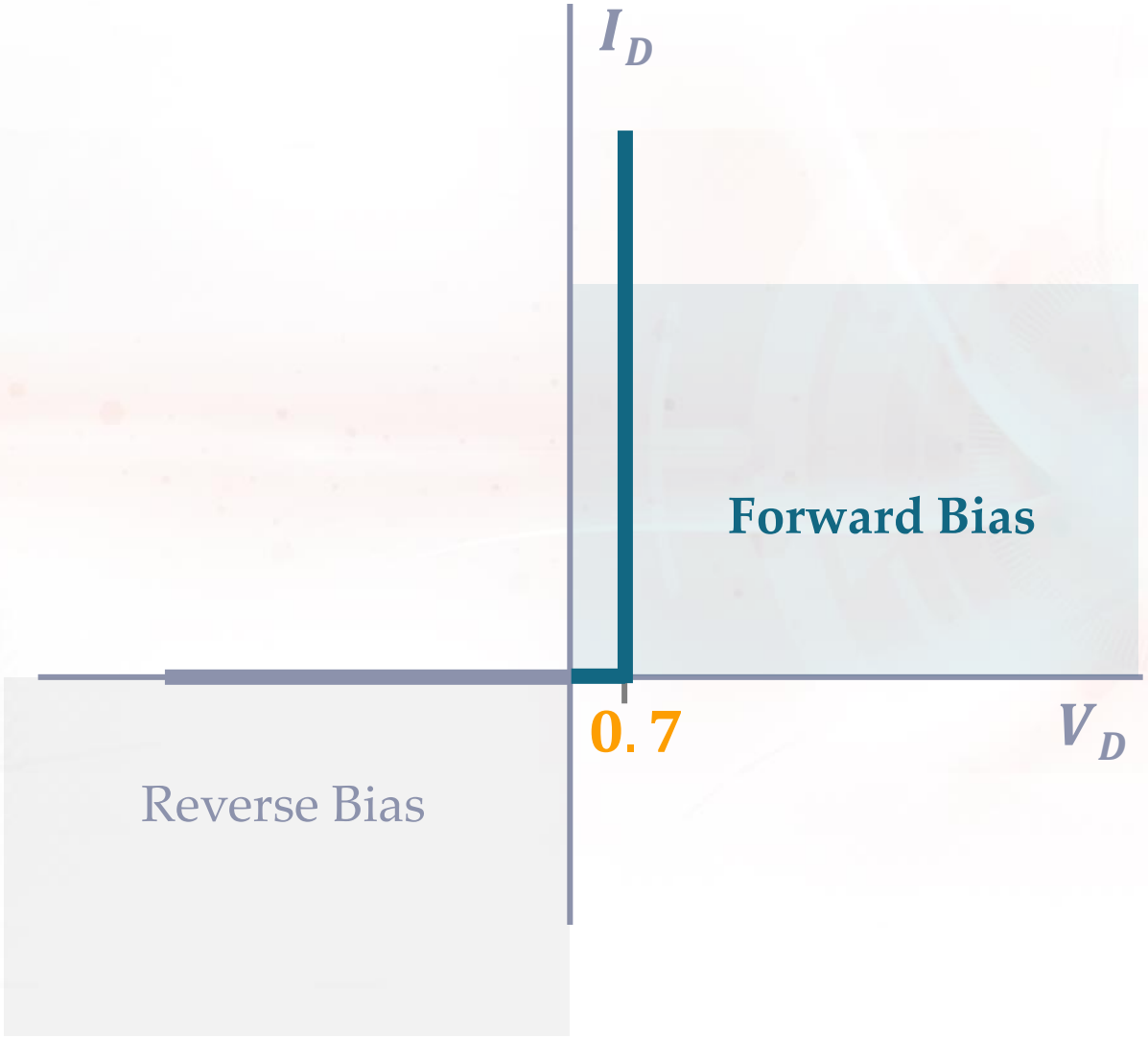


2ND APPROXIMATION

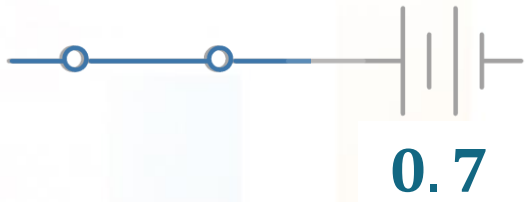
Characteristic Curve



Switch in series with a **barrier potential**

If the $\boldsymbol{V_D \geq 0.7}$, the switch is **close**

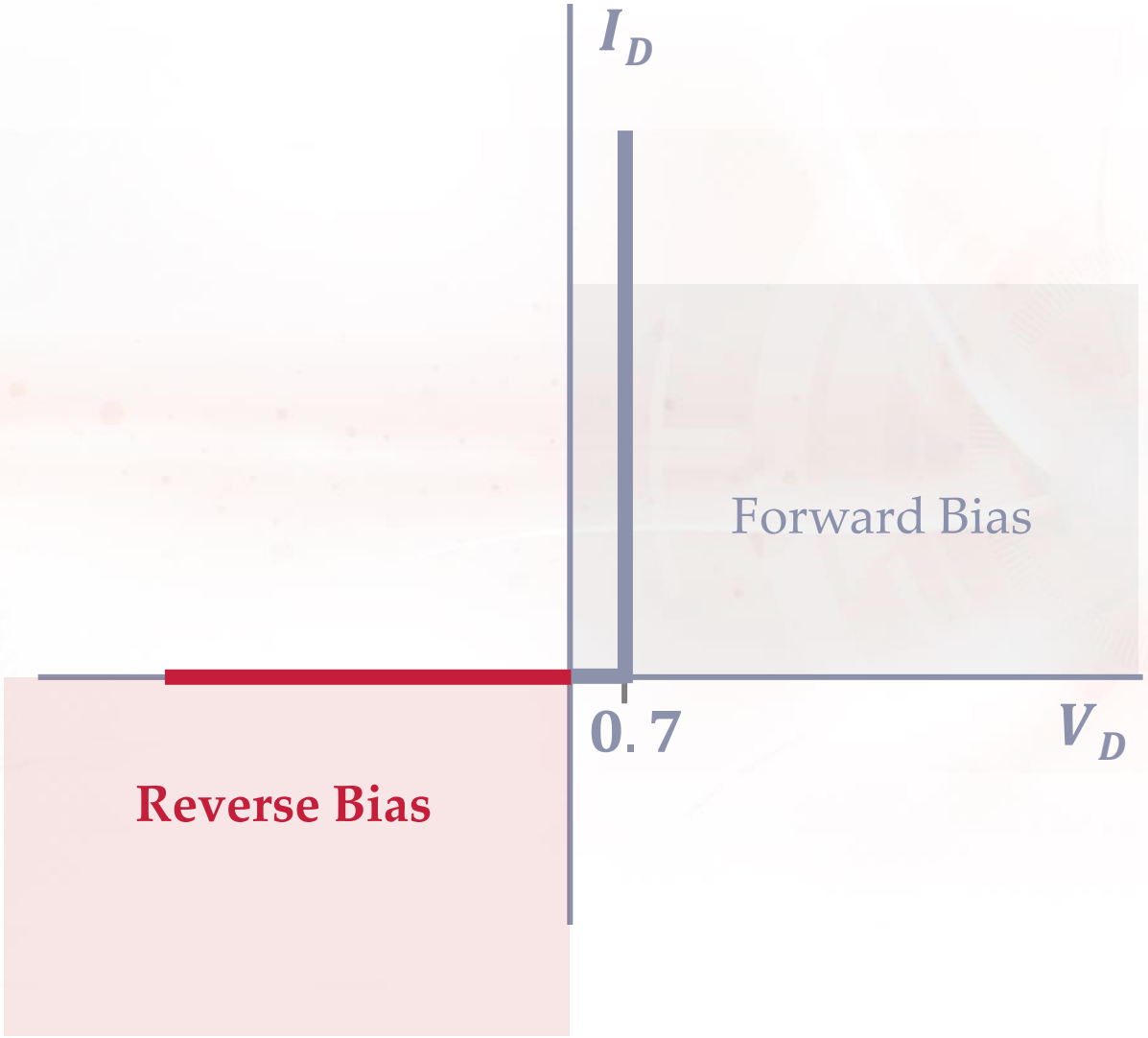
Circuit Equivalent



Forward Bias

2ND APPROXIMATION

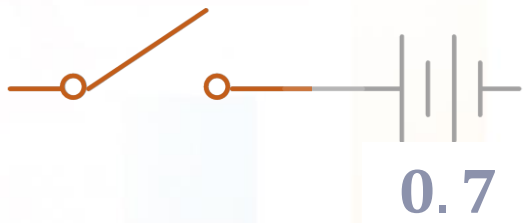
Characteristic Curve



Switch in series with a **barrier potential**:

If the $\boldsymbol{V_D < 0.7}$, the switch is **open**

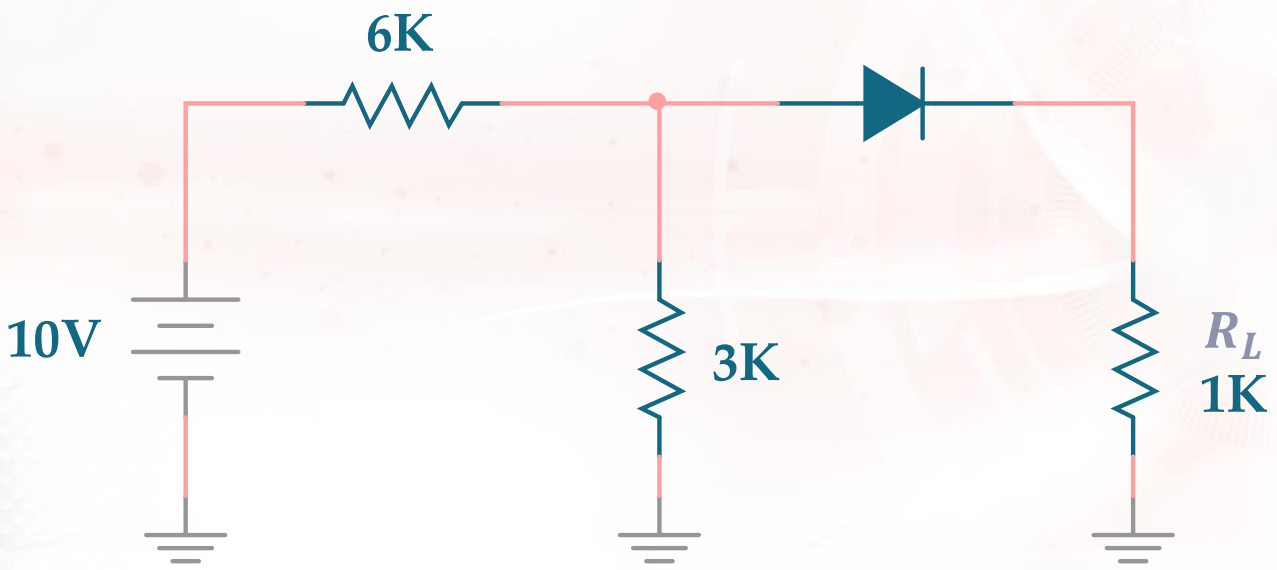
Circuit Equivalent



Reverse Bias

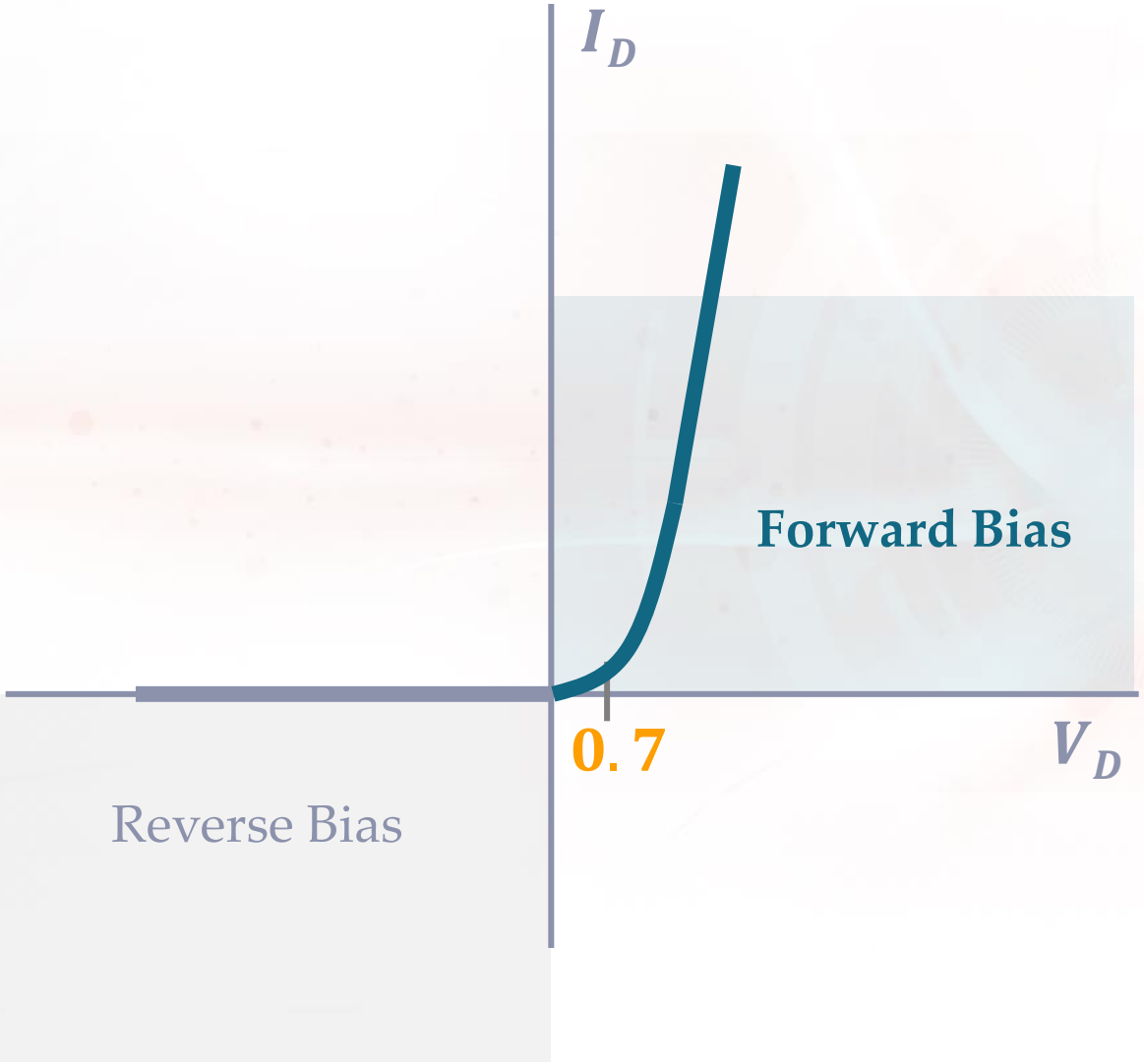
Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.

Use the **2nd approximation** diode to calculate the load **voltage** and load **current** of the given network.



3RD APPROXIMATION

Characteristic Curve

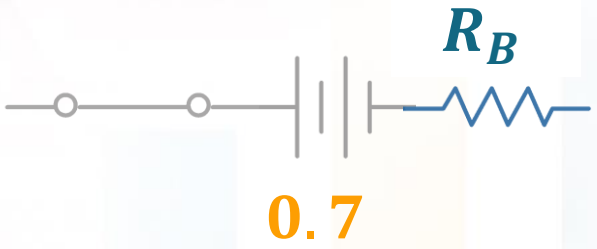


Bulk Resistance is included:

$$V_D = 0.7V + I_D R_B$$

Ignore bulk: $R_B < 0.01 R_{TH}$

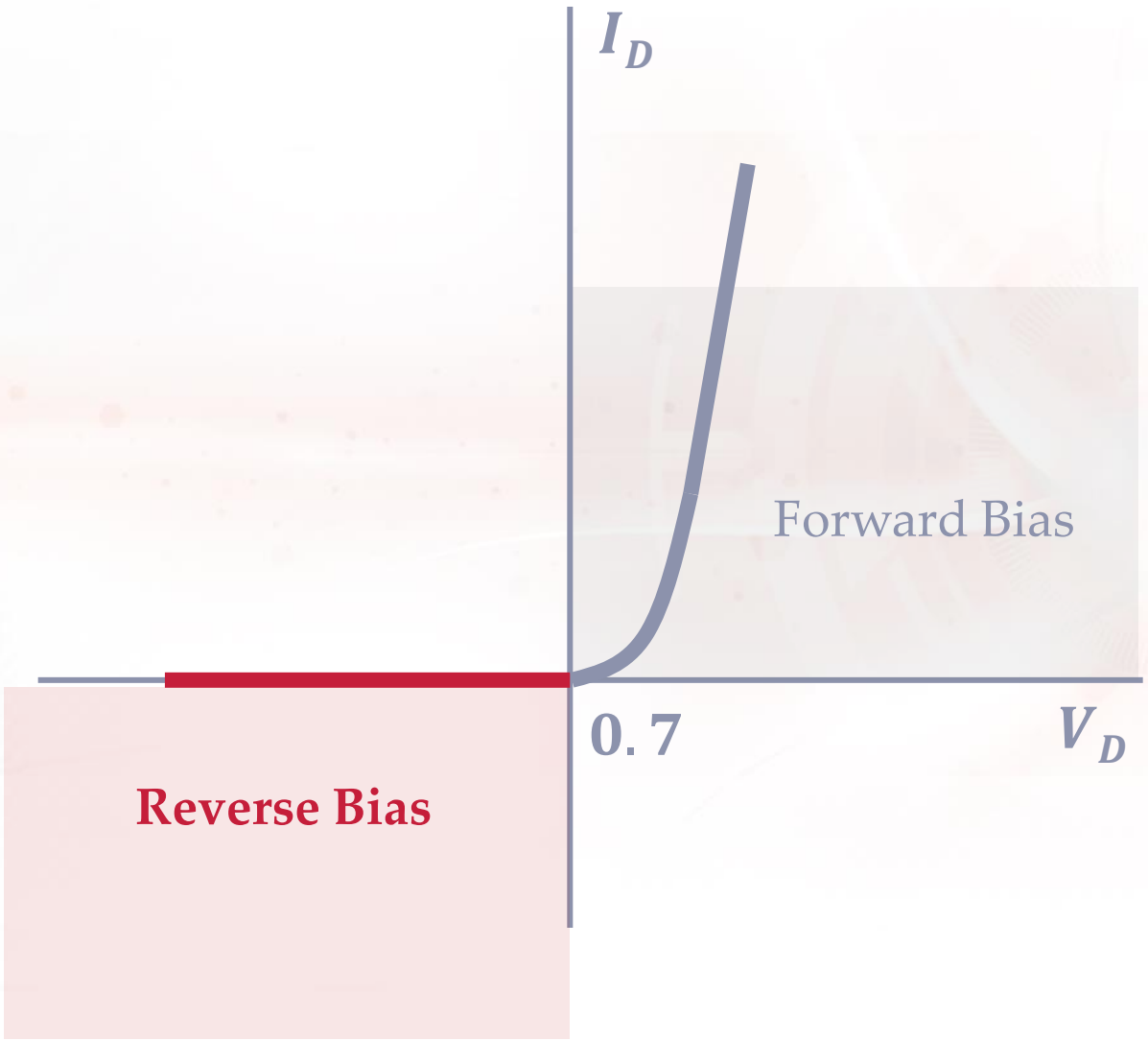
Circuit Equivalent



Forward Bias

3RD APPROXIMATION

Characteristic Curve



Bulk Resistance is included:

$V_D = 0.7V + I_D R_B$

Ignore bulk: $R_B < 0.01 R_{TH}$

Circuit Equivalent

Reverse Bias

BULK RESISTANCE

The sum of ohmic resistance of $\boldsymbol{p}$ and $\boldsymbol{n}$ region

$$\boldsymbol{R_B = R_P + R_N}$$

$$\boldsymbol{R_B < 1\Omega} \quad (typ)$$

The 1N4001 of the given network has a bulk resistance of **0.23 Ω**. What is the load **voltage**, load **current**, and **diode power**?

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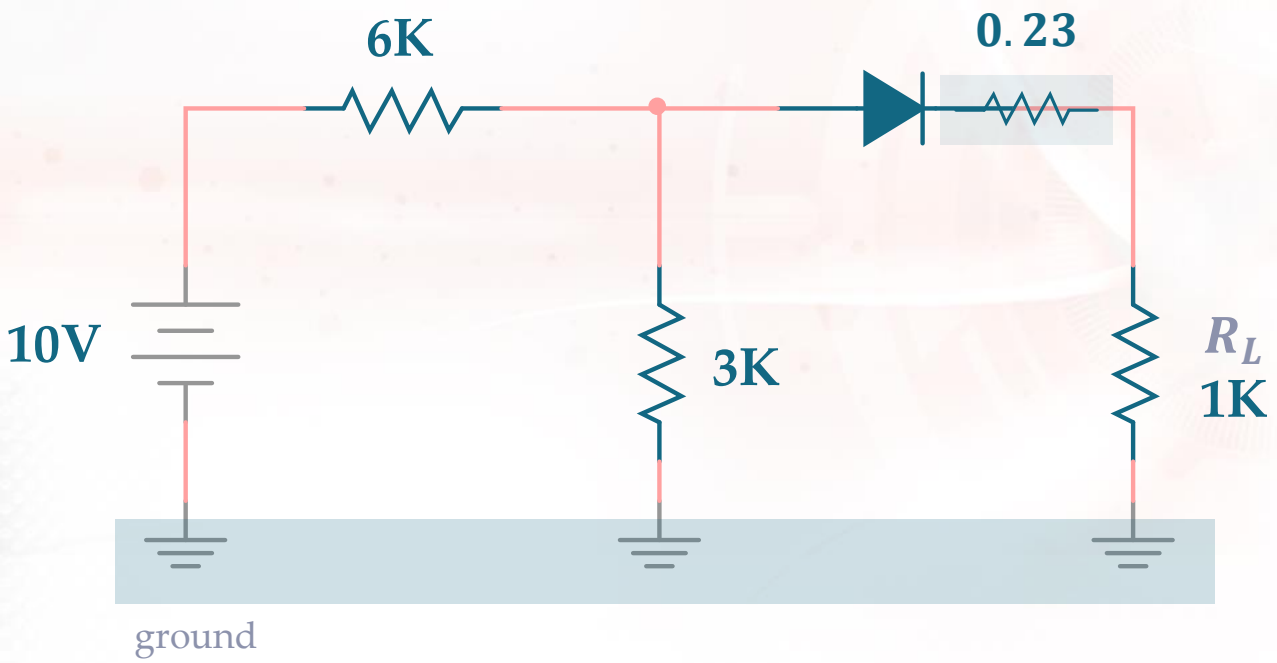
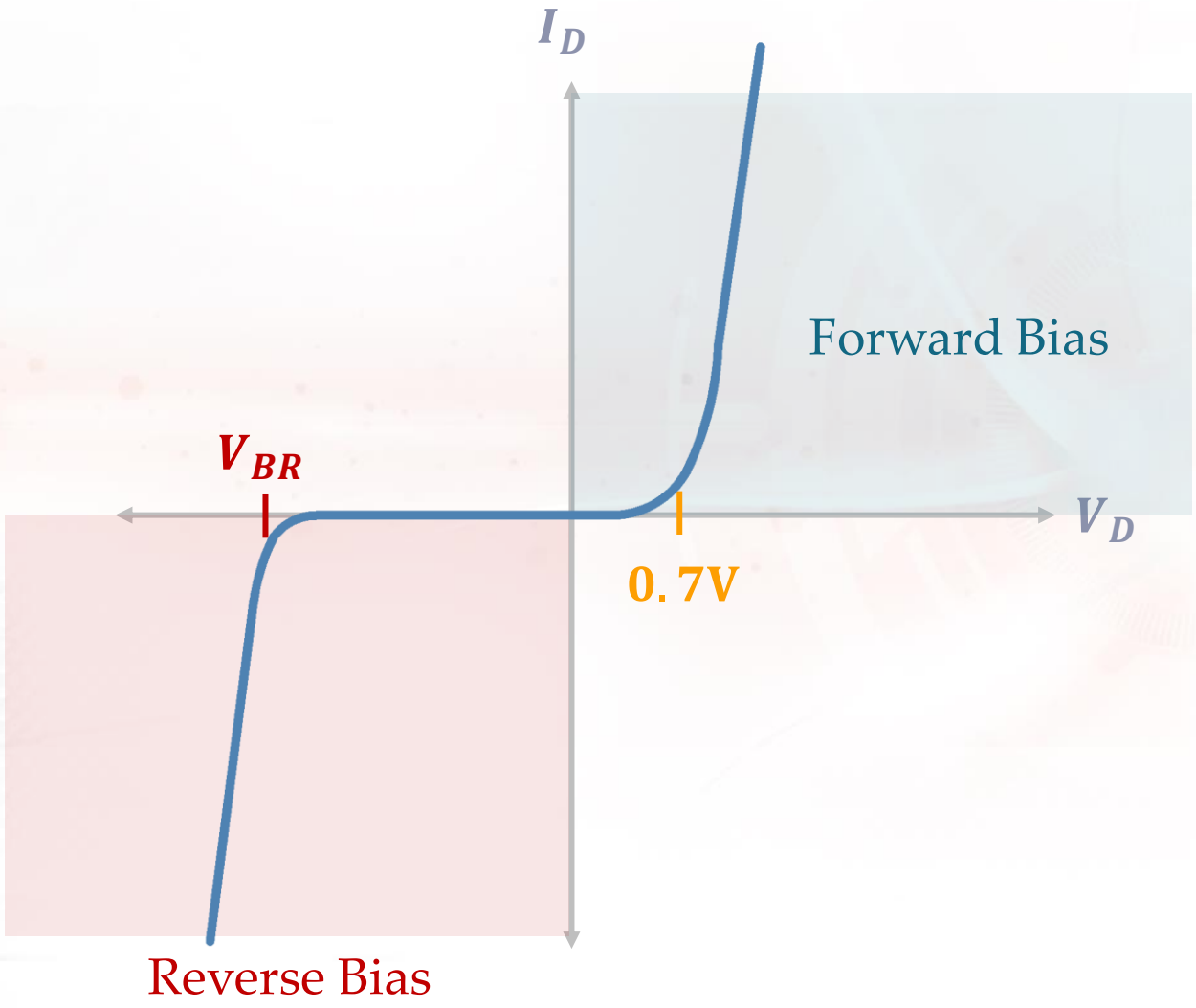


SHOCKLEY'S EQUATION

Diode Characteristic Curve



$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

where,

I_D = forward bias current

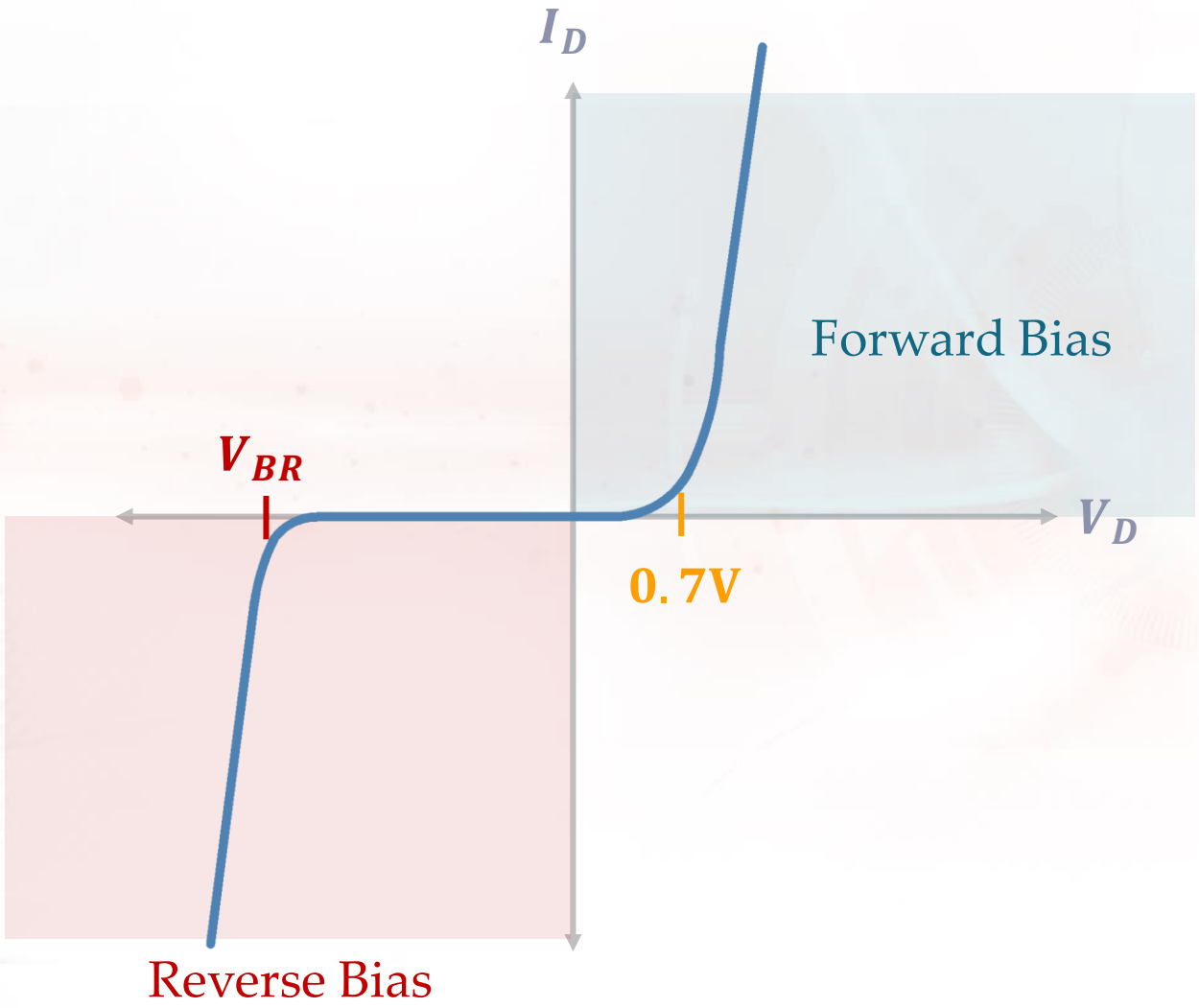
I_S = reverse saturation current

V_D = forward bias diode voltage

n = ideality factor (typ. = 1)

V_T = thermal voltage

Diode Characteristic Curve



$$V_T = \frac{kT}{q}$$

where,

k = Boltzmann's constant
$= 1.38 \times 10^{-23} J/K$

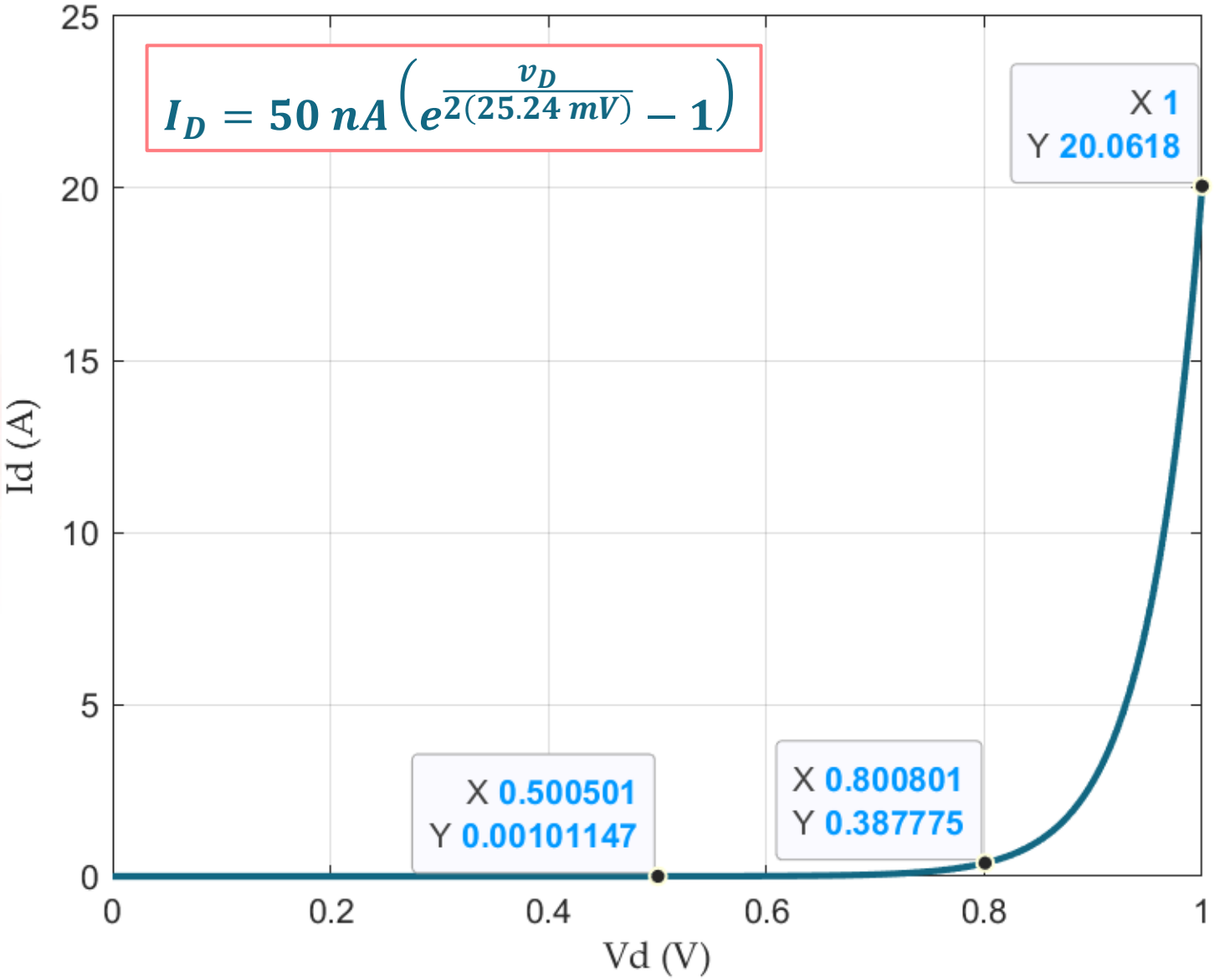
T = absolute temperature
$= \ ^\circ C + 273$

q = electronic charge
$= 1.602 \times 10^{-19} C$

EXERCISE

Determine the **diode current** at **20℃** for silicon diode with $\boldsymbol{I_s = 50nA}$, $\boldsymbol{n = 2}$ and an applied voltage of **0V, 0.5V, 0.7V,** and **1V** .

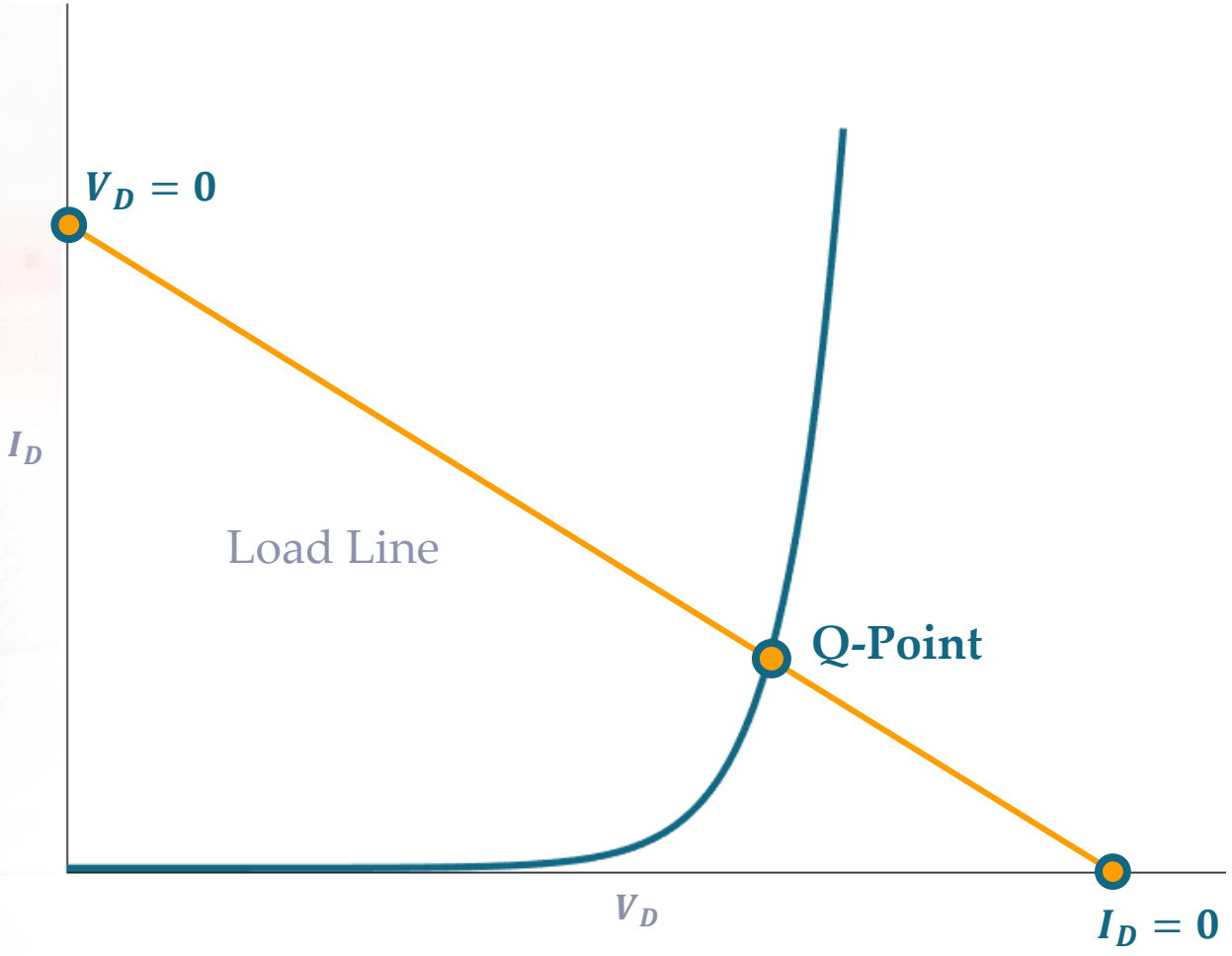
Diode Curve


$I_D = 50\ nA\left(e^{\frac{v_D}{2(25.24\ mV)}} - 1\right)$
X 1
Y 20.0618
X 0.500501
Y 0.00101147
X 0.800801
Y 0.387775
Id (A)
Vd (V)
0
5
10
15
20
25
0
0.2
0.4
0.6
0.8
1

LOAD LINE ANALYSIS

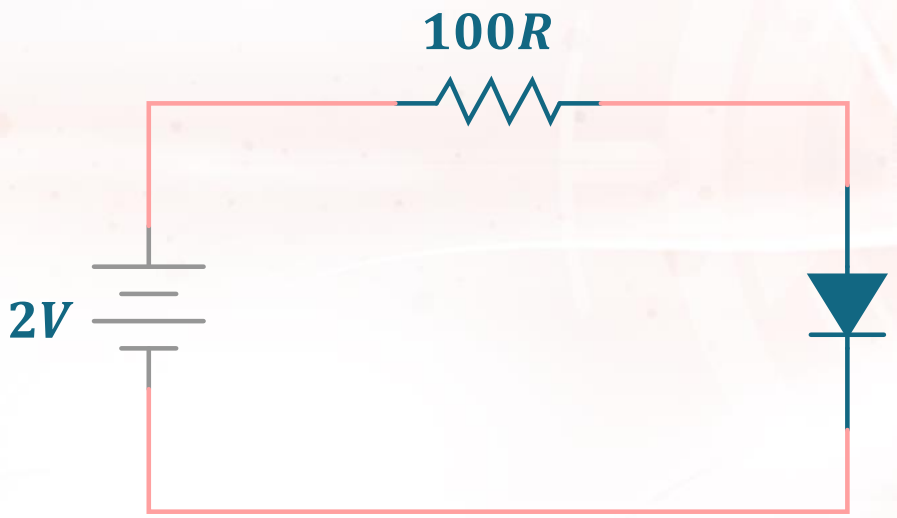
QUIESCENT POINT

Diode Characteristic Curve

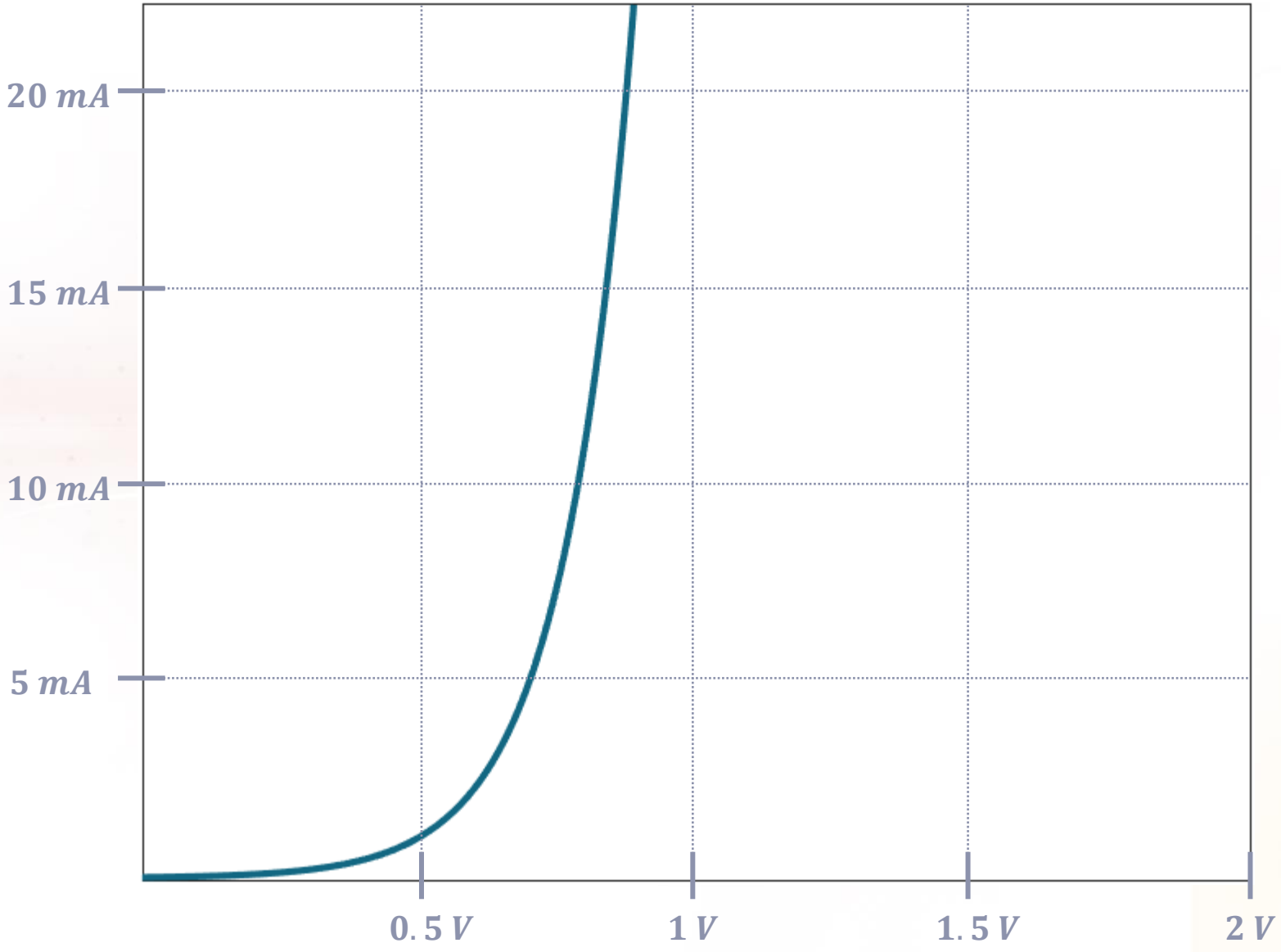


The **intersection** of the load line with the characteristic curve determines the **point of operation** of the system

Determine the point of operation, **Q-Point** of the given circuit.



Diode Curve



READING A DATA SHEET

REVERSE BREADOWN VOLTAGE

MAXIMUM RATINGS

Rating	Symbol	1N4001	1N4002	1N4003	1N4004	1N4005	1N4006	1N4007	Unit
†Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	50	100	200	400	600	800	1000	V
†Non–Repetitive Peak Reverse Voltage (halfwave, single phase, 60 Hz)	V_{RSM}	60	120	240	480	720	1000	1200	V
†RMS Reverse Voltage	$V_{R(RMS)}$	35	70	140	280	420	560	700	V
†Average Rectified Forward Current (single phase, resistive load, 60 Hz, T_A = 75°C)	I_O	1.0							A

Destructive level to avoid under all operating conditions

Safety factor of 2

MAXIMUM FORWARD CURRENT

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Level of forward current the diode **burns** out

because of excessive power dissipation

FORWARD VOLTAGE DROP

ELECTRICAL CHARACTERISTICS†

Rating	Symbol	Typ	Max	Unit
Maximum Instantaneous Forward Voltage Drop, (i_F = 1.0 Amp, T_J = 25°C)	v_F	0.93	1.1	V
Maximum Full–Cycle Average Forward Voltage Drop, (I_O = 1.0 Amp, T_L = 75°C, 1 inch leads)	$V_{F(AV)}$	–	0.8	V
Maximum Reverse Current (rated DC voltage) (T_J = 25°C) (T_J = 100°C)	I_R	 0.05 1.0	 10 50	μA
Maximum Full–Cycle Average Reverse Current, (I_O = 1.0 Amp, T_L = 75°C, 1 inch leads)	$I_{R(AV)}$	–	30	μA

The typical voltage across the diode

LABORATORY